

Of course. Here is the translation of the provided document.

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T527 PCB Design Guide

Version: 1.0

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Preface

Confidentiality Level: Secret

Overview

This document primarily introduces the key points and detailed rules for PCB design in application solutions for the T527 chip. It aims to guide customers in their design, help shorten the mass production cycle of products, improve the stability of product design, and reduce the product failure rate. At the same time, please use the core module templates released by Allwinner Technology to ensure product performance.

Chip Model

The chip model corresponding to this document is: T527

Applicable Audience

This document is mainly applicable to the following engineers:

- Product Hardware Development Engineers
- Technical Support Engineers
- PCB Layout Engineers

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Version History

Version	Date	Author/Reviser	Reviewer	Version Description
1.0	2023-11-29	AWA1798	AWA0905, AWA1651, AWA1795, KPA0570	Initial version

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1 PCB Stackup Structure Design

PCB stackup design needs to consider the impact of the distance between signals and reference planes on impedance. Combined with minimum trace width and spacing design requirements, it achieves all impedance design requirements for inner and outer layers, as well as single-ended and differential lines.

Layer Definition Design Principles:

- The layer adjacent to the main chip should be a ground plane to provide a reference plane for routing on the component side.
- All signal layers should be adjacent to a ground plane as much as possible.
- Avoid having two signal layers directly adjacent to each other.
- In principle, the stackup should adopt a symmetrical structure. Symmetry includes the thickness and type of dielectric layers, copper foil thickness, and symmetry of the pattern distribution type (large copper pour layer, circuit layer).

The following are common recommended stackup arrangements for reference.

1.1 Four-Layer Board Stackup Design

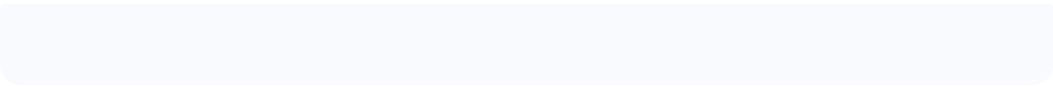
For a board thickness of 0.8mm-1.6mm, the recommended stackup structure and impedance control are as follows:

- **L1:** Main routing layer, with L2 as its reference plane.
- **L2:** Ground plane, with a full copper pour.
- **L3:** L2 and L4 as reference planes. Used for DDR area routing. For other areas, power routing is recommended.
- **L4:** Reference plane. In the DDR area, it should be a power or ground plane with a large copper pour. Used for routing in other areas.

Figure 1.1 Four-Layer PCB Stackup

(The translated table is presented below for clarity)

Layer	Type	Thickness (mil)	Dk (with Sim ZO)	Impedance spec (Ohms)	Reference layer	Width/space (mil)	Sim ZO (Ohms)	Layer definition
DDR								
1	TOP	1.6 (0.3oz+plating)	3.5-4.5	37.5±10%	2	8.1	NA	Signal
				50±10%	2	4.5	NA	
				75±10%	2	< 3mil	NA	
				75±10%	Cross-layer 3	20mil	NA	
				85±10%	2	5/5	NA	
				90±10%	2	4.4/5	NA	
	100±10%	2	4/8	NA				
	prepreg	2.5-3.2	3.5-4.5					
2	L2 GND	1.2 (1.0oz)						GND
	core	20-50	3.8-4.5					
3	L3	1.2 (1.0oz)		37.5±10%	4/2	6.7	NA	Signal
				50±10%	4/2	4	NA	
				85±10%	4/2	4.1/5	NA	
				90±10%	4/2	3.8/6	NA	
	prepreg	2.5-3.2	3.5-4.5					
	4	BOTTOM	1.6 (0.3oz+plating)	3.5-4.5	37.5±10%	3	8.1	NA
50±10%					3	4.5	NA	
75±10%					3	< 3mil	NA	
75±10%					Layer 2	20mil	NA	
85±10%					3	5/5	NA	
90±10%					3	4.4/5	NA	
100±10%					3	4/8	NA	
Board thickness:					31-63			



Note:

- The board manufacturer's adjustment of trace width should not exceed 15% of the original design, and it is recommended to be within 10%. For example, a trace designed at 4mil should not exceed 4.6mil in actual production.
- The impedance calculation results in column 1 are based on a PP thickness of 2.9mil and are for reference only.

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1.2 Six-Layer Board Stackup Design

For board thicknesses of 0.8mm/1.0mm/1.2mm, the recommended stackup structure and impedance control are as follows:

- **L1:** Routing layer, with L2 as its reference plane.
- **L2/L5:** Ground planes, with a full copper pour.
- **L3:** L2 and L4 as reference planes. Recommended for power routing. If signal routing is necessary, avoid overlapping with L4.
- **L4:** L3 and L5 as reference planes. Main routing layer.
- **L6:** Ground plane or power plane.

Figure 1.2 Six-Layer Stackup for 0.8mm/1.0mm/1.2mm Board Thickness

(The translated table is presented below for clarity)

Layer	Type	Thickness (mil)	Dk (with Sim Z0)	Impedance spec (Ohms)	Reference layer	Width/space (mil)	Sim Z0 (Ohms)
1	TOP	1.2 (0.3oz+plating)	3.5-4.5	50±10%	2	4.5	50.52
				85±10%	2	5/6	85.37
				90±10%	2	4.4/5	89.61
				100±10%	2	4/8	99.87
2	prepreg	2.5-3.2	3.5-4.5				
	GND1	0.6-1.2 (0.5/1.0oz)					
3	core	4.0	3.5-4.5				
	SIG	0.6-1.2 (0.5/1.0oz)	3.5-4.5	50±10%	2&4	4.5	50.7
4	SIG	0.6-1.2 (0.5/1.0oz)	3.5-4.5	85±10%	2&4	5/6.5	86.5
				90±10%	2&4	4.4/6.1	89.4
				100±10%	2&4	4/7	99.2
	prepreg	10-24	3.5-4.5				
5	SIG	0.6-1.2 (0.5/1.0oz)	3.5-4.5	50±10%	3&5	4.5	50.7
				85±10%	3&5	5/6.5	86.5

Layer	Type	Thickness (mil)	Dk (with Sim Z0)	Impedance spec (Ohms)	Reference layer	Width/space (mil)	Sim Z0 (Ohms)
5	core	4.0	3.5-4.5	90±10%	3&5	4.4/6.1	89.4
				100±10%	3&5	4/7	99.2
	GND2	0.6-1.2 (0.5/1.0oz)					
	prepreg	2.5-3.2	3.5-4.5				
6	BOTTOM	1.2 (0.3oz+plating)	3.5-4.5	50±10%	5	4.5	50.52
				85±10%	5	5/5	85.37
				90±10%	5	4.4/5	89.61
				100±10%	5	4/8	99.87
				Board thickness:	31.6-47.2		

Note:

- The board manufacturer's adjustment of trace width should not exceed 15% of the original design, and it is recommended to be within 10%. For example, a trace designed at 4mil should not exceed 4.6mil in actual production.
- The impedance calculation results in column 1 are based on a PP thickness of 2.9mil and are for reference only.
- When L3 and L4 both have routing in the same area (top-down view), avoid mutual overlap.

For a board thickness of 1.6mm, the recommended stackup structure and impedance control are as follows:

- **L1:** Main routing layer, with L2 as its reference plane.
- **L2/L5:** Ground planes, with a full copper pour.
- **L3:** L2 and L4 as reference planes. Recommended for routing major and minor power supplies or lower-speed signals, and should not overlap with L4 routing.
- **L4:** L3 and L5 as reference planes. Main routing layer.
- **L6:** Ground plane or power plane.

Figure 1.3 Six-Layer Stackup for 1.6mm Board Thickness

(The translated table is presented below for clarity)

Layer	Type	Thickness (mil)	Dk (with Sim Z0)	Impedance spec (Ohms)	Reference layer	Width/space (mil)	Sim Z0 (Ohms)	Layer definition
DDR								
1	TOP	1.6 (0.3oz+plating)	3.5-4.5	37.5±10%	2	8.1	NA	Signal

Layer	Type	Thickness (mil)	Dk (with Sim Z0)	Impedance spec (Ohms)	Reference layer	Width/space (mil)	Sim Z0 (Ohms)	Layer definition
2				50±10%	2	4.5	NA	
				85±10%	2	5/5	NA	
				90±10%	2	4.4/5	NA	
				100±10%	2	4/8	NA	
	prepreg	2.5-3.2	3.5-4.5					GND
	L2 GND	1.2 (1.0oz)						
	core	20-22	3.8-4.5					
3	L3	1.2 (1.0oz)	3.5-4.5	37.5±10%	2&4	7.5	NA	Signal/power
				50±10%	2&4	4.5	NA	
				85±10%	2&4	5/6.5	NA	
				90±10%	2&5	4.4/6.1	NA	
				100±10%	2&5	4/7	NA	
	prepreg	3.5-4.5	3.5-4.5					Signal
	L4	1.2 (1.0oz)	3.8-4.5	37.5±10%	3&5	7.5	NA	
				50±10%	3&5	4.5	NA	
				85±10%	3&5	5/6.5	NA	
				90±10%	3&5	4.4/6.1	NA	
				100±10%	3&5	4/7	NA	
4	core	20-22	3.8-4.5					
	L5 GND	1.2 (1.0oz)	3.5-4.5	50±10%	4&6	4	NA	
				85±10%	4&6	4.3/6.2	NA	
5	prepreg	2.5-3.2	3.5-4.5					GND
	BOTTOM	1.6 (0.3oz+plating)		37.5±10%	5	8.1	NA	
				50±10%	5	4.5	NA	
				85±10%	5	5/5	NA	
				90±10%	5	4.4/5	NA	
				100±10%	5	4/8	NA	
								Signal
				Board thickness:	63			

Note:

- The board manufacturer's adjustment of trace width should not exceed 15% of the original design, and it is recommended to be within 10%. For example, a trace designed at 4mil should not exceed 4.6mil in actual production.
 - The impedance calculation results in column 1 are based on a PP thickness of 2.9mil and are for reference only.
 - L3 and L4 are reference layers for each other. Therefore, in the same area, only one of L3 or L4 can have routing; the other layer needs to be a ground or power pour.
-

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2 SOC Fan-out

Taking a common 6-layer PCB as an example:

- The first, second, and some of the third rows of balls can be fanned out directly on the top layer (trace width 3Mil, trace spacing 3Mil).

Figure 2.1 Top Layer Routing

- L2, L5 are GND layers. Vias should be arranged neatly to leave as much routing channel space as possible. Fill in the gaps between vias where possible.

Figure 2.2 L2/L5 Layer Routing

- The small power balls and some signal balls from the third to the fifth rows are fanned out using 8/16Mil vias and routed on the L3 layer. Power trace width ≥ 9 mil, signal trace width 3mil, trace spacing 3mil. Vias should be arranged neatly to leave as much routing channel space as possible.
-

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Figure 2.3 L3 Layer Routing

Figure 2.4 L4 Layer Routing

- A small number of signal balls and small power balls from the third to the fifth rows are fanned out using 8/16Mil vias and routed on the L4 layer. Power trace width ≥ 9 mil. For different signal groups, leave as much routing channel space as possible.
 - The central power balls are fanned out using 8/16Mil vias and routed on the bottom layer. Vias should preferably use via channels to leave as much power channel space as possible.
 - VDD-CPUB requires at least 6 vias.
 - VDD-CPUL requires 5 vias.
-

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- VDD-SYS requires 4 vias.
- VDD-GPU requires 4 vias.
- VDD-DNR requires 4 vias.
- VDD-DRAML requires 4 vias.
- VDD-DRAM requires 5 vias.

- Pour copper foil in the channels as shown in the figure.

Figure 2.5 Bottom Layer Routing

4-Layer PCB

The routing on the top layer and L2 is the same as for a 6-layer PCB. L3 is the main inner routing layer; be careful not to cross over the power channels on the bottom layer. The bottom layer routing requirements are the same as for a 6-layer PCB.

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3 General Routing Recommendations

- It is recommended to calculate current capacity as 1A for 40mil copper foil and 0.5A for an 8/16Mil via. For all power vias, if space permits, add 2-3 extra vias for margin and use a large copper pour.
- Minimize corners in traces. It is recommended to use 135-degree angles instead of 90-degree angles.
- All ground guard traces should be at least 8Mil away from the signal trace; otherwise, the impedance will be affected.
- Routing should have a complete and continuous reference plane.

Figure 3.1 Incomplete Reference Plane

- The reference plane on the layer adjacent to any voided components must be a ground plane.
- When a trace changes layers, and the reference planes before and after the change are both ground planes, a corresponding ground via should be placed next to the signal via. For differential signals, the signal vias and return path vias should be placed symmetrically. For single-ended CLK signals, it is recommended to place a return path via next to the signal via.

Figure 3.2 Adding Return Path Vias

- Crosstalk from the serpentine trace itself can affect signal delay. When routing for length matching, it is recommended that the spacing $S \geq 3W$.

Figure 3.3 $S \geq 3W$

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- Differential signals require intra-pair length matching, meaning the delay difference between P and N should be as small as possible. Therefore, when a delay difference occurs between the P and N traces of a differential pair, compensate by adding a serpentine trace nearby. The dimensions of the serpentine trace require special attention and should meet the requirements shown in the figure below to reduce the impact of impedance discontinuities.

Figure 3.4 Differential Trace Length Matching Requirement

Image shows a serpentine trace with dimensions labeled A, B, C, D, E, F, G, S1, and S2.

Recommended dimensions: $A=B=C=D$, $E=F=G=3W$ (W =trace width) and $S2 < 2*S1$. The overall length of the serpentine section should be $< 2.54\text{mm}$.

- If an intra-pair length mismatch (within 300 Mil) occurs in a differential pair, compensate for it with serpentine routing as soon as possible.

Figure 3.5 Differential Trace Length Matching Requirement 2

Image shows length matching being performed close to where the mismatch occurs.

- When routing serpentine traces within a differential pair, ensure the spacing remains constant.

Figure 3.6 Differential Trace Routing Spacing

Image shows that the spacing within the serpentine section should be kept the same as the original inter-pair spacing.

- When a differential pair changes layers, the vias should be kept symmetrical.
-

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Figure 3.7 Maintain Symmetry for Layer-Changing Vias

Image shows a "Preferred" symmetrical via placement and an "Avoid" non-symmetrical placement for a differential pair changing layers.

3.1 High-Speed Routing Requirements

For signals with a rate exceeding 3Gbps/lane, such as USB3.0, PCIe2.1, and HDMI, please follow the recommendations below:

3.1.1 Differential Via Recommendations

For differential trace vias, it is recommended that one end of the via be on the top layer, and the other end be on an inner layer near the bottom or on the bottom layer itself. Furthermore, only the top, bottom, and breakout layers of the via should have pads, as shown in the right part of the figure below. This technique is often referred to as back-drilling or using controlled-depth vias to remove the unused portion (stub) of the via barrel.

Figure 3.8 Trace Pad Handling

Image compares a via with pads on all layers to a via with pads only on the top, bottom, and breakout layers.

The via dimensions for interface differential pairs should be optimized through simulation based on the actual stackup, and the anti-pad should be cleared on all layers.

Below are reference via dimensions based on a reference stackup:

- R_Drill = 4 Mil (drilling radius)
 - R_Pad = 8 Mil (via pad radius)
 - D1: Center-to-center spacing of differential vias: 22 Mil
 - D2: Anti-pad dimension from the surface to the bottom layer: 18 Mil
 - D3: Center-to-center spacing between a signal via and a return path ground via: 21-26 Mil
-

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Figure 3.9 Anti-Pad Dimensions

Image shows the dimensions D1, D2, and D3 for a differential via pair with accompanying ground vias.

3.1.2 Coupling Capacitor Optimization Recommendations

Void the copper on the second layer (ground plane) directly underneath the coupling capacitor pads. The voided area should be 2 mils larger than the pad in both length and width.

Figure 3.10 Voiding Area for Coupling Capacitors

Image shows a capacitor with a void in the ground plane underneath it, with the text "比焊盘大2mil" (2mil larger than the pad).

3.1.3 ESD Component Optimization Recommendations

Void the copper on the second layer (ground plane) directly underneath the ESD component pads. The voided area should be 2 mils larger than the pads in both length and width. The GND vias for the large central pad should be as symmetrical as possible.

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Figure 3.11 Voiding Area for ESD Components

3.1.4 Connector Optimization Recommendations

Void the copper on the second layer (ground plane) directly underneath the connector pads. The voided area should be 2 mils larger than the pad in both length and width.

Figure 3.12 Voiding Area for Connector Pads

Image shows a connector with a void in the ground plane underneath its pads, with the text "比焊盘大2mil" (2mil larger than the pad).

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3.2 Length Matching Requirements Table

Table 3.13 High-Speed Line Length Matching Requirements Table

Name	Characteristic Impedance ohm (±10%)	Length Matching Requirement	Spacing	VIA Count
USB2.0	90	Intra-pair < 50mil, Total length ≤ 6000mil	≥ 10mil from others	< 2
USB3.0	100	USB3.0 TX, RX intra-pair ≤ 5mil, Total length ≤ 5000mil	> 20mil (inter-pair & from others)	< 2
PCIE2.1	100	Intra-pair ≤ 5mil, Total length ≤ 5000mil	> 20mil (inter-pair & from others)	≤ 2
MIPI-DSI	100	Intra-pair ≤ 5mil, Inter-pair < 160mil (1080P)	> 15mil	≤ 2
MIPI-CSI	100	Intra-pair ≤ 5mil, Inter-pair ≤ 300mil	≥ 15mil	CLK ≤ 2
EDP	100	Intra-pair ≤ 5mil, Total length ≤ 3000mil	> 20mil	< 2
HDMI1.4	100	Intra-pair ≤ 5mil, Total length ≤ 3000mil	> 15mil	< 2
HDMI2.0	100	Intra-pair ≤ 5mil, Total length ≤ 2000mil	≥ 15mil	< 2
LCD/LVDS	100	Intra-pair < 20mil, Inter-pair < 450mil (720P); < 180mil (1080P)	> 10mil	< 2
Internal EPHY	100	Intra-pair < 20mil, Inter-pair < 200mil	≥ 10mil	< 2

Table 3.14 Low-Speed Line Length Matching Requirements Table

Name	Characteristic Impedance ohm ($\pm 10\%$)	Length and Length Matching Requirement	Spacing	VIA Count
NAND	50	D0~D7, RE, WE matched to DQS $\leq$ 300mil, Total length $\leq$ 2000mil	> 9mil	D0~D7 should have the same number of vias
EMMC	50	D0~D7, DS, CMD matched to CLK $\leq$ 300mil, Total length $\leq$ 2000mil	> 9mil	D0~D7 should have the same number of vias
SDIO	50	D0~D3, CMD matched to CLK $\leq$ 300mil	> 9mil	D0~D3 should have the same number of vias
SD CARD	50	D0~D3, CMD matched to CLK < 500mil	> 9mil	D0~D3 should have the same number of vias

4 Layout Design for Major Modules

4.1 Crystal Oscillator

- When placing the crystal oscillator, do not put it at the edge of the PCB. Place it as close to the chip as possible to avoid overly long crystal traces, which should not exceed 600 Mil.
- The crystal oscillator must be placed on the same side as the SOC to avoid frequency shifts caused by layer-changing vias.
- The matching capacitors for the crystal oscillator must be placed close to it.
- The placement of the crystal oscillator needs to be far away from heat sources, as high temperatures can also affect its frequency deviation.
- The periphery of the crystal oscillator and its traces, as well as adjacent layers, should be shielded and protected by GND. The crystal traces need to be well-protected by GND and kept away from sensitive signals.

Figure 4.1 TOP Layer Routing

Figure 4.2 L2 Layer Routing

- The X32KFOUT clock is used by the WIFI module. Its trace must be shielded by a ground guard on the same layer. Keep it away from high-current and high-speed traces.

4.2 AP-RESET and AP-NMI

It is recommended to follow these principles for the layout of AP-RESET and AP-NMI pins:

- The pull-up resistor for the SOC reset signal should be close to the SOC. The reset signal should be guarded by ground on both sides, and the capacitor to ground should be placed close to the SOC to improve ESD performance.

Figure 4.3 AP-RESET Component Placement and Routing

- The AP-NMI signal should be guarded by ground on both sides, and the capacitor to ground should be placed close to the SOC to improve ESD performance.

Figure 4.4 AP-NMI Component Placement and Routing

4.3 AUDIO CODEC

- The grounding capacitors and resistors for AVCC, VRP, VRA1, VRA2, and AGND should be placed sequentially close to the main controller.
- The decoupling capacitors on CPVDD, CPVEE, VPP, and VEE should be placed close to the chip.
- The grounding resistor for HPOUTFB should be placed close to the headphone jack.
- The microphone should be placed near the edge of the circuit board, ≥ 200 Mil away from RF, PA, and switching power supplies.
- ESD devices must be placed close to the MIC. Traces coming from the MIC must first pass through the ESD devices before connecting to other components.
- AVCC trace width ≥ 10 Mil; VRP, VRA1, VRA2 trace width ≥ 10 Mil; keep away from high-speed interference signals.

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- AGND should have a copper pour area. The copper pour width should be ≥ 20 Mil, and the AGND grounding resistor should have ≥ 2 vias connecting to the GND plane.

Figure 4.5 AGND Routing

- The grounding capacitors and resistors for AVCC, VRA1, VRA2, VRP, and HPCOMFB should connect to the AGND copper pour.
- The trace widths for CPVEE and VEE should be ≥ 10 Mil, and the length should be ≤ 300 Mil. They should form a star connection at the 10uF capacitor, as shown in the figure below.

Figure 4.6 Star Connection

- HPOUTL, HPOUTFB, and HPOUTR should be routed in parallel and guarded by ground. HPOUTL and HPOUTR should be on the sides with HPOUTFB in the middle, and the entire group should be guarded by ground. The trace width should be 10 Mil. Keep the traces and vias away from high-speed signals and clock signals.
- The grounding resistor for HPOUTFB should be close to the headphone jack pad. The trace should first go to the headphone jack and then to the grounding resistor. The signal must be tapped from the headphone jack end to the MIC end.

Figure 4.7 HP Signal Group Guarded by Ground

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Figure 4.8 HPOUTFB Line Must Be Tapped from the Headphone Jack

Image caption: The grounding resistor should be close to the headphone jack pad. The trace should first pass through the headphone jack and then to the grounding resistor.

- MICXP, MICXN should be routed as a pseudo-differential pair, with a trace width of at least 4 Mil and spacing of at least 4 Mil, and guarded by ground.
- MBIAS, HBIAS should be routed in parallel with MICXP/MICXN, with a trace width of 10 Mil and guarded by ground.

- LINEOUTLP, LINEOUTLN should be routed as a pseudo-differential pair, with a trace width of at least 4 Mil and spacing of at least 4 Mil, and guarded by ground.
- LINEOUTRP, LINEOURLN should be routed as a pseudo-differential pair, with a trace width of at least 4 Mil and spacing of at least 4 Mil, and guarded by ground.

Figure 4.9 LINE Guarded by Ground

4.4 EMMC

- All decoupling capacitors for VCCQ, VCC, and VDDI should be placed close to the EMMC.
- The series resistor for the CLK signal should be placed close to the main controller, as shown in the figure below. The trace distance between the series resistor and the main controller's CLK pin should be ≤ 300 Mil.

Figure 4.10 EMMC Series Resistor and Main Controller CLK

- The pull-down resistor for the DS signal should be placed close to the EMMC. The stub length introduced by the pull-down resistor should be ≤ 200 Mil.

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Figure 4.11 Stub Length of DS Signal Pull-Down Resistor

Image caption: Stub length

- The trace impedance should be 50 ohm.
- The trace length between the EMMC and the main controller should be ≤ 2000 Mil, with a routing spacing of ≥ 9 Mil.
- Except for Reset, ensure that all signal traces have a complete reference plane, as shown in the diagram below:

Figure 4.12 EMMC Routing Reference Plane

- The CLK and DS signals should be guarded by ground, with the guard trace connected to the GND plane via vias. If ground guarding is not possible, maintain a trace spacing of ≥ 12 Mil.
- DO~D7, DS, and CMD should have their lengths matched to CLK within ≤ 300 Mil. The number of vias used on DO~D7 should be as consistent as possible.
- VCC and VCCQ trace widths should be ≥ 12 Mil, or use a copper pour instead of traces for power. If there are vias on the power lines, the number of vias should be no less than 2 to avoid current limiting by the vias affecting the power supply.
- Avoid routing near high-frequency signals.
- EMMC NC/RFU and other reserved pins should all be left floating. Do not connect these signals to power, ground, or other EMMC signals for routing convenience. If routing is indeed difficult, the EMMC PCB footprint can be appropriately modified to remove some NC/RFU balls.
- If you expect the EMMC to operate at a higher frequency, it is recommended to use only the EMMC to ensure a point-to-point connection between the main controller's IO and the EMMC. If a dual layout for NAND/EMMC is used, adopt a daisy-chain routing method, making the EMMC the endpoint of the trace to minimize the length of stubs.

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Figure 4.13 EMMC Routing Reference

4.5 NAND

- All decoupling capacitors for VCCQ and VCC should be placed close to the NAND.
- The series resistor for the RE signal should be placed close to the main controller. The trace distance between the series resistor and the main controller should be ≤ 300 Mil.
- The trace impedance should be 50 ohm.
- The trace length between the NAND and the main controller should be ≤ 2000 Mil; trace spacing should be ≥ 9 Mil.
- Maintain a trace spacing of ≥ 12 Mil for RE, WE, and DQS.
- Length matching should be controlled to ≤ 300 Mil.
- DO~D7, RE, and WE should be length-matched to DQS within ≤ 300 Mil.
- Ensure a complete reference plane.
- VCC and VCCQ trace widths should be ≥ 12 Mil.
- If you expect the NAND to operate at a higher frequency, it is recommended to use only the NAND to ensure a point-to-point connection between the main controller's IO and the NAND.

4.6 SPI

- The SPI flash should be placed close to the main controller.
- The series resistor for the CLK signal should be placed close to the main controller. The trace distance between the series resistor and the main controller should be ≤ 300 Mil.
- The series resistors for MOSI, MISO, WP, and CS should be placed close to the FLASH. The trace distance between the series resistors and the FLASH should be ≤ 300 Mil.

Figure 4.14 Place Series Resistor Close to the Main Controller

- The trace impedance should be 50 ohm; routing spacing should be ≥ 9 Mil; ensure a complete trace reference plane.
-

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- The CLK signal should be guarded by ground or kept at a distance of ≥ 12 Mil from other signal lines.
- VCC trace width should be ≥ 12 Mil.
- Avoid routing near other high-speed, high-frequency signals.

Figure 4.15 CLK Signal with Ground Guarding

4.7 SMART CARD

- The series resistor for CLK should be placed close to the main controller. The trace distance between the series resistor and the main controller's CLK pin should be ≤ 300 Mil.
- ESD devices should be placed close to the interface/card socket. The traces from the card socket pins should first connect to the ESD devices before connecting to other components.
- The resistor and capacitor network on the VDD line should be placed close to the interface/card socket.
- Do not place through-hole components on the back of the card socket to prevent their pins from interfering with card insertion and removal.

Figure 4.16 SMART CARD Key Component Placement

- The trace impedance should be 50 ohm.
- Trace spacing should be ≥ 9 Mil.

- DO~D3 and CMD should be length-matched to CLK within < 500 Mil.
 - The CLK signal should be guarded by ground, with the guard trace connected to the GND plane via vias. If ground guarding is not possible, maintain a trace spacing of ≥ 12 Mil.
 - VDD trace width should be ≥ 12 Mil.
 - Avoid routing near high-frequency signals.
-

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4.8 WIFI/BT

- Place the module as close as possible to the antenna or antenna connector. Keep it away from modules that can easily generate interference, such as power supplies, DDR, LCD circuits, cameras, motors, and speakers.
- Lay out the matching capacitors and resistors for the antenna feedline according to the schematic. The antenna feedline should be routed smoothly, as short as possible, without branches, without vias, and with few corners.
- The antenna feedline should have a controlled impedance of 50 ohm, be guarded by ground on both sides, and have numerous GND vias along the path. To increase trace width and reduce loss on multi-layer boards, it is common to void the adjacent layer under the feedline, with the 3rd layer needing to be a complete reference ground. The return path for the reference layer's EPAD must be complete and not split. The trace width should be the same as the capacitor and resistor pad widths as much as possible.

Figure 4.17 WIFI/BT Antenna Feedline Routing 1

Figure 4.18 WIFI/BT Antenna Reference Layer EPAD Return Path Complete

- If using a PCB trace as an antenna, ensure the area near the antenna trace is completely clear. The keep-out area should be greater than 50mm². The antenna body itself should be at least 1cm away from any surrounding metal.

Figure 4.19 WIFI/BT On-board Antenna

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4.9 SDIO

- The 33K resistor for CLK should be placed close to the SOC side.
- Figure 4.20 Placement of CLK 33K Resistor**
- The trace impedance should be 50 ohm.
 - Trace spacing should be ≥ 9 Mil.
 - The CLK signal should be guarded by ground, with the guard trace connected to the GND plane via vias. If ground guarding is not possible, maintain a trace spacing of ≥ 12 Mil.
 - DO~D3 and CMD should have their lengths matched to CLK within ≤ 300 Mil.
 - Avoid routing near high-frequency signals.

Figure 4.21 SDIO Routing

4.10 EPHY

- The transformer should be placed close to the RJ45 jack.
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Figure 4.22 Transformer Placed Near RJ45 Jack

- The Ethernet LED lights should be placed close to the jack.

Figure 4.23 Ethernet LED Lights

- Do not pour copper on any layer underneath the RJ45 and transformer. The keep-out distance should be 40-80 Mil.

Figure 4.24 No Copper Pour on All Layers Underneath RJ45 and Transformer

Image labels show 40-80mil keep-out areas.

4.11 NCSI

- The capacitor to ground for PCLK should be near the main controller, and the series resistor should be near the module.
 - The capacitor to ground for MCLK should be near the module, and the series resistor should be near the main controller.
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- The capacitor to ground for Hsync should be near the main controller.
- The series resistors for Vsync, Hsync, and Data should be near the module.
- The filter capacitors for AVDD, IOVDD, and DVDD should be placed near the module.
- MCLK and PCLK each require ground guarding with GND vias. If PCB space is limited and complete signal line guarding cannot be guaranteed, ensure that there are no other traces within a space of ≤ 15 Mil from these signal lines.
- Ensure minimal layer changes during routing (MCLK, PCLK layer-changing vias ≤ 2).
- The spacing for other traces should be greater than 8 Mil.
- DATA, HSYNC, and VSYNC should be length-matched as a group to PCLK within 500 Mil. PCLK should not be routed in a serpentine fashion just to achieve length matching.
- The ground guard for PCLK needs to extend to the sensor and IC ends. Do not place CSI_DATA lines on either side of PCLK. Keep it away from high-speed switching signal lines, or isolate it with a ground trace. If the PCLK and DATA balls are adjacent, route PCLK and DATA to different layers at the breakout.

4.12 EDP

- The ESD devices and DC-blocking capacitors on the EDP signal lines should be placed close to the EDP connector, as shown in the right figure below:

Figure 4.25 Placement of ESD Devices and DC-Blocking Capacitors

- EDP signal lines should be routed differentially, with a required differential impedance of 100 ohm.
 - The spacing between differential pairs should be ≥ 20 Mil.
 - The distance from a differential pair to other signals should be ≥ 20 Mil, with a ground trace separating it from high-speed signals.
 - The EDP PWR supply pin trace width should be greater than or equal to 20 Mil.
 - The EDP signal line length should be ≤ 2000 Mil. The intra-pair length matching error should be ≤ 5 Mil, and the inter-pair length matching error should be ≤ 100 Mil.
-

4.13 LCD/LVDS

- The 104 (0.1uF) filter capacitor for VCC-PD must be placed as close as possible to the IC side.

Figure 4.26 VCC-PD Placement

4.13.1 LCD

- The CLK line must be guarded by ground, and attention should be paid to placing vias on the guard trace. If ground guarding is not possible, maintain a distance of ≥ 12 Mil between this signal line and other signals.
- The reference plane for the LCD lines must be complete. If possible, the Dx data lines can be grouped in pairs.

Figure 4.27 Recommended Data Routing

4.13.2 LVDS

- Impedance requirement: single-ended 50 ohm, differential 100 ohm.
- Backlight circuit requirement: The trace width for the nets PS, VLED+, and VLED- should be 20 Mil or more.
- The distance from the CLK line to other signals should be ≥ 12 Mil.
- For 1080P: Intra-pair length difference within 10 Mil, inter-pair length difference within 180 Mil.
- For 720P: Intra-pair length difference within 20 Mil, inter-pair length difference within 450 Mil.

4.14 HDMI

- ESD devices should be close to the HDMI connector. The distance from the ESD device to the HDMI connector should be less than or equal to 500 mil. See the figure below:

Figure 4.28 ESD Devices Placed Near HDMI Connector

- The signal lines should be routed differentially, with a required differential impedance of 100 ohm.
- **HDMI 1.4:** HDMI signal line length ≤ 3000 mil. Do not deliberately use "serpentine" routing to achieve length matching. The HDMI differential intra-pair length matching error should be < 5 mil, and the inter-pair length matching error should be ≤ 200 mil.
- **HDMI 2.0 and above:** HDMI signal line length ≤ 2000 mil. Do not deliberately use "serpentine" routing to achieve length matching. The HDMI differential intra-pair length matching error should be ≤ 5 mil, and the inter-pair length matching error should be ≤ 200 mil.

4.15 MIPI DSI

- Impedance requirement: single-ended 50 ohm, differential 100 ohm. Number of layer-changing vias ≤ 2 . When changing layers with vias, DP/DN must change layers with vias simultaneously. The CLK and DATA lines of the same group should follow the same topology, meaning the number of layer changes and vias must be consistent.
- Differential constraints: Intra-pair length difference within 10 Mil, inter-pair length difference within 160 Mil.

Figure 4.29 MIPI DSI Routing

4.16 MIPI CSI

- The filter capacitors for AFVCC, AVDD, IOVDD, and DVDD should be placed near the module.
 - MIPI differential impedance is 100 ohm. Number of layer-changing vias ≤ 2 . When changing layers with vias, DP/DN must change layers with vias simultaneously, and more ground vias should be added nearby. The CLK and DATA lines of the same group should follow the same topology, meaning the number of layer changes and vias must be consistent.
 - MIPI traces should maintain a continuous reference plane, preferably a ground layer. If this condition cannot be met, it is essential to ensure that the width of the reference plane can reach 16 mil.
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Figure 4.30 MIPI CSI Routing

- To reduce interference between differential pairs, separate each pair with a ground trace, or maintain a spacing of ≥ 15 mil.
- Intra-pair length matching within 5 mil, inter-pair length matching ≤ 300 mil (the smaller, the better).

4.17 PCIE 2.1

- The decoupling and filter capacitors for VCC should be placed close to the IC.
- The decoupling capacitor for the PCIE interface should be placed close to the interface's power PIN.
- The series capacitors for PCIE_TX should be placed close to the IC side, not exceeding 500 Mil.

Figure 4.31 PCIE Component Placement

- The series capacitors for PCIE_RX should be placed close to the interface side, not exceeding 500 Mil.
- The differential impedance is 100 ohm.
- Differential pairs should preferably be routed on the Top layer, followed by the Bottom layer and inner layers near the bottom (if vias are used, ensure the via stub is short).
- P/N should always be routed as a parallel differential pair. Ensure the signal trace length is controlled within 5000 Mil, the number of vias does not exceed 2, and the differential trace length error is 5 Mil.

4.18 USB

4.18.1 USB2.0

- The trace width for the power supply VCC3V3-USB should be ≥ 20 Mil, and for USB-5V, it should be ≥ 40 Mil.
-

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Figure 4.32 VBUS PIN Routing

- The decoupling and filter capacitors for VCC3V3-USB need to be placed close to the IC.
- The decoupling capacitor for the USB SOCKET should be placed close to the SOCKET's VBUS PIN.
- ESD devices should be placed close to the connector end, and the routing length between the connector and the ESD device should be ≤ 500 mil.

Figure 4.33 ESD Device Placement

- The USB differential impedance is 90 ohm. The trace length should be ≤ 6000 Mil, DM/DP length matching should be ≤ 50 Mil, and the number of vias should not exceed 2.

- It is recommended to prioritize the routing of DM/DP, maintain a spacing of greater than 10 Mil from other signals, and avoid routing under components or crossing other signals.

4.18.2 USB3.0

- ESD devices should be placed close to the connector end, and the routing length between the connector and the ESD device should be ≤ 500 mil.
- The series capacitors for SS_TX should be placed close to the IC side.
- The differential impedance is 100 ohm.
- SS differential pairs should preferably be routed on the Top layer, followed by the Bottom layer and inner layers near the bottom (if vias are used, ensure the via stub is short).
- Ensure that the adjacent layer for the traces has a continuous and complete reference plane, and that the reference plane is not split.
- The trace length should be controlled within 5000 mil, and the number of vias should not exceed 2.
- SS_TXP/N and SS_RXP/N should always be routed as a parallel differential pair. The P/N signal length matching should be controlled within 5 Mil. The length matching adjustment for SS_RX should be close to the interface connector; the adjustment for SS_TX should be close to the SOC side.

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Figure 4.34 USB3.0 Component Placement and Routing

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5 Power Supply

5.1 AXP717/AXP323 Power Supply Design

- The PMIC and major components like inductors and capacitors are recommended to be placed on the same layer. For designs where space around the PMIC is particularly tight, large components like inductors and capacitors can be placed directly underneath the PMIC on the bottom layer. It is recommended that DCDC output capacitors be placed perpendicular to the inductors to reduce interference from the inductor's LX signal during ripple testing.
- If an inductor is connected to the PMIC, while ensuring the capacitor's position, place it as close as possible to the PMIC's LX pin. The trace width should be sufficient for the current requirement.
- Do not route sensitive signals within 40 Mil of the LX power plane area.

Figure 5.1 AXP717 Component Placement

Figure 5.2 AXP323 Component Placement

- The capacitor for VREF should be as close as possible to the pin. The grounding point of the capacitor should be as far away as possible from switching nodes to avoid interference.
- The communication signals between the system and the PMIC are also switching signals and should be kept away from sensitive signals.

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- For the voltage feedback line, after the output voltage is filtered by a capacitor, tap the feedback point right next to the capacitor and route it to the PMIC with a 4~10 mil trace. It is best not to route it under the inductor, under the AC current path, or adjacent to switching signals like CLK. For CPU/SYS/DRAM power supplies, use remote sensing. If the SOC has a feedback pin, connect to the feedback pin. If there is no feedback pin, connect to the two terminals of the remote sense capacitor. The feedback line should run alongside its power plane to the load.
- The pad at the bottom of the PMIC should have multiple vias connecting to the ground plane. Use a solid connection on the ground layer to facilitate heat dissipation.
- For USB charging, the DP and DM resistors should be placed no more than 700 mil from the PMIC.
- Vxx-xxFB should use remote sensing. The feedback line should avoid sensitive clock signals like CLK, stay away from sensitive signal vias, and run alongside its power plane to the load.

Figure 5.3 USB Charging Resistor Placement and Routing

5.2 Discrete Power Supply Layout Design

It is recommended to follow these principles for power supply layout:

- DCDC, LDO, and major components like inductors and capacitors are recommended to be placed on the same layer.
- The input filter capacitor should be as close as possible to the power input pin. If vias are used, there should be no fewer than 3 vias.
- The placement of the DCDC output capacitor and inductor must ensure the shortest possible loop area to reduce loop inductance and minimize interference from the inductor's LX signal during ripple testing.

Figure 5.4 BUCK Topology

Image shows a BUCK converter circuit diagram with DC path (直流路径) and AC path (交流路径) highlighted.

- If an inductor is connected to the DCDC, while ensuring the capacitor's position, place it as close as possible to the DCDC's LX pin. The trace diameter should meet the current requirement. It is best for the DCDC input to have a power plane.

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- For the voltage feedback line, after the output voltage is filtered by a capacitor, tap the feedback point right next to the capacitor and route it to the DCDC with a 4~10 mil trace.
- The parallel trace section of the feedback line on the TOP layer with the LX node should be as short as possible. It is best not to route it under the inductor, under the AC current path, or adjacent to switching signals like CLK. The ground of the voltage divider resistors should be as close as possible to the DCDC's ground.

Figure 5.5 Feedback Line Routing

Image label: 反馈线 (Feedback line)

- Routing and ground pours are prohibited underneath the DCDC inductor.

5.3 Power Supply Simulation Requirements:

IR-drop:

- The DC voltage drop at the input of VDD-CPUB, VDD-CPUL, VDD-SYS, VDD-GPU, and VDD-DNR should be $< \pm 2\%$.

AC Impedance Requirement:

- For VDD-CPUB, VDD-CPUL, VDD-SYS, VDD-GPU, and VDD-DNR, the impedance should be < 0.03 ohm below 10 MHz, and < 0.4 ohm from 10 MHz to 100 MHz.

Figure 5.6 Power Supply AC Impedance Curve

Image shows an Impedance vs. Frequency graph.

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